MASOUD ABDOLLAHI, Ph.D.

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Research Scientist with a Ph.D. and 4+ years of industry experience building and deploying machine learning systems at scale. Expert in deep learning (CNNs, RNNs/LSTMs, Transformers), foundation models, vision-language models, and multimodal AI; applied to real manufacturing environments with measurable product impact. Author of 28+ peer-reviewed publications (700+ citations), 6 patent filings (1 published), and a track record of leading research agendas end-to-end from prototype to production. Experienced contributor to the broader research community through publications, peer review, and cross-institutional collaboration.

RESEARCH EXPERTISE

- Machine & Deep Learning (CNN, RNN/LSTM, Transformer)
- Foundation Models & Vision-Language Models
- Multimodal & Self-Supervised Learning
- Computer Vision & Pose Estimation
- Time Series Classification & Sensor Fusion
- Generative AI & LLM-powered Systems
- Human Motion Analysis & NLP
- Data Mining & Experimental Design
- Python, PyTorch, TensorFlow, Azure OpenAI, HuggingFace

PROFESSIONAL EXPERIENCE

Senior R&D Scientist · Hitachi America, Ltd. Feb. 2023 – Present

- Promoted to Senior R&D Scientist (Jun 2026); lead multiple R&D projects and own a research roadmap.
- Led end-to-end research and deployment of a Vision-Language Model (VLM) + generative AI system that cut a critical manufacturing bottleneck by **25%** at a U.S. Hitachi facility, from research concept to full production deployment.
- Designed and shipped deep learning models (CNN, RNN-LSTM, Transformer) and classical ML pipelines on real automotive OEM datasets to solve manufacturing quality and warranty prediction challenges.
- Built multimodal perception systems combining wearable sensors and computer vision (pose estimation, action recognition), achieving **>90% accuracy** across real production environments.
- Accumulated **500+ hours/year** of on-site factory data collection, bridging lab research and industrial-scale deployment.
- Inventor on **6 patent filings in AI/ML, generative AI, and sensing for industrial applications** (1 published: US 2026/0198641 A1; 2 under USPTO review; 3 approved for filing).
- Collaborated cross-functionally with engineering, IP/legal, and leadership teams to define research roadmap and technical strategy.

Research Engineer III (Part-time) · Oregon Health & Science University Feb. 2024 – Present

- Developing sensor-based deep learning models to analyze freezing of gait and fall risk in Parkinson's disease from wearable-derived data, alongside clinicians and medical domain experts.

Graduate Research Assistant · Biomechanics & Ergonomics Lab, RIT Aug. 2017 – Dec. 2023

- Designed and led a novel data-driven fall risk assessment model for stroke survivors using 8 IMU sensors and clinical cognitive dual-tasking, resulting in multiple top-journal publications and a defended Ph.D. dissertation.
- Developed camera-based fall risk assessment, smartphone fatigue detection (SVM; 91% accuracy), and LBP patient classification (SVM/MLP; 75%): three independent research agendas led from hypothesis to publication.
- Contributed to NIH proposal writing (R01, R03, R21) and external grants, building research strategy alongside faculty and clinical collaborators.
- Led clinical data collection at Unity Hospital with patients with neurological conditions; mentored junior students and teaching assistants across 10+ semesters.

R&D Scientist Intern · Aster Pet Sciences Jun. – Aug. 2021

- Led small team to design, collect data, and ship a real-time IMU-based dog activity recognition system (SVM; >70% accuracy). First full ownership of an AI research-to-product pipeline.

SELECTED PUBLICATIONS

28+ peer-reviewed publications (700+ citations) spanning machine learning, deep learning, computer vision, and human motion. Full list: Google Scholar

- Abdollahi et al. "360° turning in individuals with mild to moderate Parkinson's disease: a clustering analysis of segmental coordination strategies." *Journal of NeuroEngineering and Rehabilitation*, 2026. DOI
- Vidyaratne, Kumar, Zhou, Abdollahi et al. "Generative AI framework for procedural knowledge extraction and training content generation in manufacturing." *ASME MSEC*, 2026. DOI
- Abdollahi et al. "Smart wearable insoles in industrial environments: A systematic review." *Applied Ergonomics*, 2024. PDF
- Abdollahi et al. "Fall risk assessment in stroke survivors: a machine learning model using detailed motion data from common clinical tests and motor-cognitive dual-tasking." *Sensors* 24(3), 2024. PDF
- Jahangiri, Abdollahi et al. "An inpatient fall risk assessment tool: Application of machine learning models on intrinsic and extrinsic risk factors." *Machine Learning with Applications*, 2024. PDF
- Jahangiri, Abdollahi et al. "A machine learning model to predict heart failure readmission: toward optimal feature set." *Frontiers in Artificial Intelligence*, 2024. PDF
- Abdollahi et al. "A systematic review of fall risk factors in stroke survivors." *Frontiers in Bioengineering and Biotechnology*, 2022. PDF
- Abdollahi et al. "Kinematic Analysis of 360° Turning in Stroke Survivors Using Wearable Motion Sensors." *Sensors* 22(1), 2022. PDF
- Abdollahi et al. "Using a Motion Sensor to Categorize Nonspecific Low Back Pain Patients: A Machine Learning Approach." *Sensors* 20(12), 2020. PDF
- Karvekar*, Abdollahi* et al. "Smartphone-based human fatigue level detection using machine learning approaches." *Ergonomics*, 2021. (*equal contribution) PDF

Active peer reviewer: Medical Engineering & Physics, Human-centric Computing & Info Sciences, IEEE Open Journal of Computer Society, Frontiers in Bioengineering & Biotechnology, Ergonomics, Pain Medicine, The Journal of Supercomputing (2017–present).

PATENTS

- Zhou, Q., & Abdollahi, M. Smart Sole Architecture for Measuring Ground Reaction Force and Method for Unifying Coordinates for Feet Positions and Gestures. U.S. Patent Application Publication US 2026/0198641 A1. Published Jul. 16, 2026.
- Abdollahi M. et al. Agentic AI & generative AI system for industrial operations. U.S. Patent Application. Under USPTO Review. Hitachi America, Ltd. (2025).
- Abdollahi M. et al. Sensor-based human performance assessment system. U.S. Patent Application. Under USPTO Review. Hitachi America, Ltd. (2026).
- Abdollahi M. et al. (3 additional applications approved for filing, in preparation: agentic AI, generative AI, sensor-based human performance assessment & training). Hitachi America, Ltd. IP & Legal Team (2026).

EDUCATION

Ph.D. in Mechanical & Industrial Engineering · Rochester Institute of Technology Aug. 2017 – Dec. 2023

Dissertation: A Comprehensive Machine Learning Approach to Assess Fall Risk in Stroke Survivors: Integrating Common Clinical Tests, Motor-Cognitive Dual-Tasks, Wearable Motion Sensors, and Detailed Motion Analysis

M.Sc. in Mechanical Engineering · Sharif University of Technology Sep. 2012 – Jun. 2015

B.Sc. in Mechanical Engineering · Semnan University Sep. 2007 – Sep. 2011

HONORS & RECOGNITION

- Winner, Hitachi America R&D AI Data Challenge (day-ahead grid price forecasting, 243 buses, 30 labeled days), 2026
- 3rd Prize, Hitachi Global Technology Award (global, company-wide R&D recognition), 2025
- Ranked 1st in Machine Learning Competition, Dept. of Computer Engineering, RIT (~60 students, mostly CS/CE), 2018
- 1 of 10 global travel grants awarded for the Autumn School on Movement Science, Berlin, Germany, 2016
- Ranked 1st among M.Sc. students in Biomechanics, Sharif University of Technology, 2015